

Introducción

Este proyecto consiste en el diseño e implementación de una infraestructura de red empresarial distribuida por provincias utilizando routers Cisco, switches administrables y protocolos de enrutamiento dinámico.

La topología cubre cinco provincias geográficas, cada una administrada por un router de borde dedicado (R4-R8), conectadas al core a través de enlaces WAN seriales redundantes (R2 y R3). Adicionalmente se incorporó un segmento de servidor web separado (R9) con dirección pública independiente.

Tecnologías Implementadas

- OSPF Área 0 único con autenticación MD5 en todos los enlaces
- Inter-VLAN Routing mediante Router-on-Stick (subinterfaces 802.1Q)
- DHCP automático por VLAN en cada router de provincia
- NAT Overload (PAT) en R1 hacia Internet
- SSH v2 exclusivo — Telnet completamente deshabilitado
- Port-Security con sticky MAC y acción shutdown
- ACLs extendidas y estándar para control de acceso y seguridad
- VLSM con redes /30 en WAN y /24 en LANs
- VLAN administrativa independiente 172.16.99.0/27
- Servidor web en zona DMZ-like administrada por R9

Descripción de la Topología

WAN VLSM COMPLETO

Redes Públicas (WAN hacia ISP)

Se utilizaron dos bloques de IP pública para los enlaces hacia el ISP:

Enlace	Red	Máscara	Router A (ISP)	Router B
ISP ↔ R1	210.10.10.0/30	255.255.255.252	210.10.10.1	210.10.10.2 (R1)
ISP ↔ R9	210.10.11.0/30	255.255.255.252	210.10.11.1	210.10.11.2 (R9)

Red WAN Privada CORE 10.0.0.0/24

Todos los enlaces internos usan /30 (2 hosts útiles, máscara 255.255.255.252):

Enlace	Red	Router A	Router B
R1 ↔ R2	10.0.0.4/30	10.0.0.5	10.0.0.6
R1 ↔ R3	10.0.0.8/30	10.0.0.9	10.0.0.10
R2 ↔ R3	10.0.0.12/30	10.0.0.13	10.0.0.14

Redes LAN Privadas por Provincia

R4

Enlace	Red	IP
R2 ↔ R4	10.0.0.16/30	R2=.17 R4=.18
R3 ↔ R4	10.0.0.20/30	R3=.21 R4=.22

R5

Enlace	Red	IP
R2 ↔ R5	10.0.0.24/30	R2=.25 R5=.26
R3 ↔ R5	10.0.0.28/30	R3=.29 R5=.30

R6

Enlace	Red	IP
R2 ↔ R6	10.0.0.32/30	R2=.33 R6=.34
R3 ↔ R6	10.0.0.36/30	R3=.37 R6=.38

R7

Enlace	Red	IP
R2 ↔ R7	10.0.0.40/30	R2=.41 R7=.42
R3 ↔ R7	10.0.0.44/30	R3=.45 R7=.46

R8

Enlace	Red	IP
R2 ↔ R8	10.0.0.48/30	R2=.49 R8=.50
R3 ↔ R8	10.0.0.52/30	R3=.53 R8=.54

VLANs PRIVADAS

R4

VLAN	Red	Gateway
VLAN 10	192.168.10.0/24	192.168.10.1
VLAN 20	192.168.20.0/24	192.168.20.1

R5

VLAN	Red	Gateway
VLAN 30	192.168.30.0/24	192.168.30.1

VLAN	Red	Gateway
VLAN 40	192.168.40.0/24	192.168.40.1

R6

VLAN	Red	Gateway
VLAN 50	192.168.50.0/24	192.168.50.1
VLAN 60	192.168.60.0/24	192.168.60.1
VLAN 70	192.168.70.0/24	192.168.70.1
VLAN 80	192.168.80.0/24	192.168.80.1

R7

VLAN	Red	Gateway
VLAN 90	192.168.90.0/24	192.168.90.1
VLAN 100	192.168.100.0/24	192.168.100.1

R8

VLAN	Red	Gateway
VLAN 110	192.168.110.0/24	192.168.110.1
VLAN 120	192.168.120.0/24	192.168.120.1

R9

server	Red	Gateway
R9	192.168.200.0/24	192.168.200.1
SERVER	192.168.200.2	192.168.200.1

VLAN ADMINISTRATIVA

Red dedicada exclusivamente para la administración remota de todos los dispositivos de la infraestructura. Máscara: 255.255.255.224 | Hosts disponibles: 30

Esta sí debe ir aparte. 172.16.99.0/27

Ubicación / Provincia	Dispositivo	IP Administrativa	Default Gateway (Router Local)	Función del Dispositivo
Provincia 1	R4-Prov1	172.16.99.1	N/A (Es Gateway)	Gateway de Provincia 1
	SW0	172.16.99.2	172.16.99.1	Switch de Acceso (VLAN 10/20)
	SW1	172.16.99.3	172.16.99.1	Switch de Acceso (VLAN 10/20)
Provincia 2	R5-Prov2	172.16.99.4	N/A (Es Gateway)	Gateway de Provincia 2
	SW2	172.16.99.5	172.16.99.4	Switch de Acceso (VLAN 30/40)
	SW3	172.16.99.6	172.16.99.4	Switch de Acceso (VLAN 30/40)
Provincia 3	R6-Prov3	172.16.99.7	N/A (Es Gateway)	Gateway de Provincia 3
	SW4	172.16.99.8	172.16.99.7	Switch de Acceso (VLAN 50/60)
	SW5	172.16.99.9	172.16.99.7	Switch de Acceso (VLAN 70/80)
	SW6	172.16.99.10	172.16.99.7	Switch de Distribución Local
Provincia 4	R7-Prov4	172.16.99.11	N/A (Es Gateway)	Gateway de Provincia 4
	SW7	172.16.99.12	172.16.99.11	Switch de Acceso (VLAN 90)
	SW8	172.16.99.13	172.16.99.11	Switch de Acceso (VLAN 100)
Provincia 5	R8-Prov5	172.16.99.14	N/A (Es Gateway)	Gateway de Provincia 5

Ubicación / Provincia	Dispositivo	IP Administrativa	Default Gateway (Router Local)	Función del Dispositivo
	SW10	172.16.99.15	172.16.99.14	Switch de Distribución Local
	SW9	172.16.99.16	172.16.99.14	Switch de Acceso (VLAN 110)
	SW11	172.16.99.17	172.16.99.14	Switch de Acceso (VLAN 120)

VLANs de usuarios

VLAN	Nombre	Red	Hosts
10	TI	192.168.10.0/24	254
20	RRHH	192.168.20.0/24	254
30	contabilidad	192.168.30.0/24	254
40	marketing	192.168.40.0/24	254
50	ventas	192.168.50.0/24	254
60	finanzas	192.168.60.0/24	254
70	logística	192.168.70.0/24	254
80	calidad	192.168.80.0/24	254
90	compras	192.168.90.0/24	254
100	almacen	192.168.100.0/24	254
110	asistente	192.168.110.0/24	254
120	CRS	192.168.120.0/24	254

Tabla de ACL y NAT

Dispositivo	ACL	Tipo	Función	Aplicada en
R4, R5, R6, R7, R8	ACL_WAN_IN	Extendida	Permitir OSPF, bloquear Telnet y permitir tráfico interno	Interfaces WAN seriales inbound
R1, R2, R3	ACL_CORE_IN	Extendida	Permitir OSPF, bloquear Telnet y permitir tráfico interno del core	Interfaces seriales del core inbound
R4, R5, R6, R7, R8	ACL_VTY_ACCESS	Estándar	Permitir acceso SSH solo desde VLAN administrativa	Líneas VTY 0 4
SW1–SW11	ACL_VTY	Estándar	Permitir administración SSH solo desde red administrativa	Líneas VTY 0 4
R7	ACL_PING_LIMIT	Extendida	Limitar tráfico ICMP	Creada pero no aplicada
R7	ACL_NO_SPOOF	Estándar	Bloquear direcciones spoofing 127.0.0.0/8	Creada pero no aplicada
R1	NAT_POOL	Estándar	Definir redes internas permitidas para NAT	Utilizada por NAT overload

Tecnologías Implementadas

OSPF — Open Shortest Path First

```
router ospf 1
```

```
router-id X.X.X.X
```

```
area 0 authentication message-digest
```

```
network 10.0.0.X 0.0.0.3 area 0
```

```
default-information originate ← Solo en R1
```

```
interface SerialX/X/X
```

```
ip ospf message-digest-key 1 md5 cisco123
```

```
ip ospf authentication message-digest
```

Proceso: OSPF 1 — Un único área (Area 0)

Autenticación: MD5 en todos los enlaces seriales

Router IDs: 1.1.1.1 (R1) hasta 8.8.8.8 (R8)

Propagación: R1 redistribuye ruta por defecto hacia toda la red

Router-on-Stick

```
interface GigabitEthernet0/1
```

```
no ip address
```

```
no shutdown
```

```
interface GigabitEthernet0/1.10
```

```
encapsulation dot1Q 10
```

```
ip address 192.168.10.1 255.255.255.0
```

```
interface GigabitEthernet0/1.999
```

```
encapsulation dot1Q 999 native
```

```
no ip address
```

DHCP

ip dhcp excluded-address 192.168.10.1

ip dhcp pool VLAN10

network 192.168.10.0 255.255.255.0

default-router 192.168.10.1

dns-server 8.8.8.8

NAT Overload (PAT) — R1

ip access-list standard NAT_POOL

permit 192.168.0.0 0.0.255.255

permit 172.16.99.0 0.0.0.31

ip nat inside source list NAT_POOL interface Serial0/0/0 overload

interface Serial0/0/0

ip nat outside

interface Serial0/0/1

ip nat inside

Hardening General

Cifrado de contraseñas

username admin privilege 15 secret cisco123

enable secret cisco123

service password-encryption

SSH v2 exclusivo

ip domain-name empresa.local

crypto key generate rsa general-keys modulus 2048

ip ssh version 2

ip ssh time-out 60

ip ssh authentication-retries 3

Protección anti fuerza bruta

login block-for 120 attempts 3 within 60

Sesiones con timeout automático

```
line vty 0 4
transport input ssh
exec-timeout 5 0
access-class ACL_VTY_ACCESS in
```

DNS deshabilitado

```
no ip domain-lookup
```

Banner de advertencia

```
banner motd # ADVERTENCIA: Acceso solo personal autorizado. #
```

Port-Security en Switches

```
interface FastEthernet0/X
switchport port-security maximum 2
switchport port-security violation shutdown
switchport port-security
switchport port-security mac-address sticky
spanning-tree portfast
```

VLAN Hopping Prevention

```
! VLAN nativa cambiada de 1 a 999
vlan 999
name UNUSED_NATIVE
interface f0/1
switchport trunk native vlan 999
switchport nonegotiate    Desactiva DTP
```

Script completo

Configuración de la provincia 1

Provincia 1-----

R4-Provincia1 (OSPF Area 0 - Router on Stick)

enable

configure terminal

hostname R4

no ip domain-lookup

banner motd # ADVERTENCIA: R4-Provincia1 - Acceso solo personal autorizado. #

username admin privilege 15 secret cisco123

enable secret cisco123

service password-encryption

ip domain-name empresa.local

crypto key generate rsa modulus 2048

ip ssh version 2

ip ssh time-out 60

ip ssh authentication-retries 3

login block-for 120 attempts 3 within 60

! ===== DHCP EXCLUDE =====

ip dhcp excluded-address 192.168.10.1

ip dhcp excluded-address 192.168.20.1

ip dhcp excluded-address 172.16.99.1

! ===== DHCP POOLS =====

```
ip dhcp pool VLAN10
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
dns-server 8.8.8.8
```

```
ip dhcp pool VLAN20
network 192.168.20.0 255.255.255.0
default-router 192.168.20.1
dns-server 8.8.8.8
```

```
! ===== INTERFACE WAN 1 =====
```

```
interface Serial0/0/0
description ENLACE WAN a R2
ip address 10.0.0.18 255.255.255.252
no shutdown
```

```
! ===== INTERFACE WAN 2 =====
```

```
interface Serial0/1/1
description ENLACE WAN R3 BACKUP
ip address 10.0.0.22 255.255.255.252
no shutdown
```

```
! ===== ROUTER ON STICK =====
```

```
interface GigabitEthernet0/1
description TRUNK AL SWITCH
no ip address
no shutdown
interface GigabitEthernet0/1.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
```

```
interface GigabitEthernet0/1.20
```

```
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
! VLAN ADMINISTRATIVA-----
```

```
interface GigabitEthernet0/1.99
encapsulation dot1Q 99
ip address 172.16.99.1 255.255.255.224
```

```
interface GigabitEthernet0/1.999
encapsulation dot1Q 999 native
no ip address
```

```
! ===== OSPF CON MD5 =====
```

```
router ospf 1
router-id 4.4.4.4
area 0 authentication message-digest
network 10.0.0.16 0.0.0.3 area 0
network 10.0.0.20 0.0.0.3 area 0
network 192.168.10.0 0.0.0.255 area 0
network 192.168.20.0 0.0.0.255 area 0
network 172.16.99.0 0.0.0.31 area 0
```

```
interface Serial0/0/0
ip ospf message-digest-key 1 md5 cisco123
ip ospf authentication message-digest
```

```
interface Serial0/1/1
ip ospf message-digest-key 1 md5 cisco123
```

```
ip ospf authentication message-digest
```

```
! ===== RUTA ESTATICA HACIA INTERNET =====
```

```
ip route 0.0.0.0 0.0.0.0 10.0.0.17
```

```
ip route 0.0.0.0 0.0.0.0 10.0.0.21 5
```

```
! ===== ACL DE SEGURIDAD =====
```

```
ip access-list extended ACL_R4
```

```
remark PERMITIR OSPF
```

```
permit ospf any any
```

```
remark BLOQUEAR BROADCAST
```

```
deny ip any host 255.255.255.255
```

```
remark BLOQUEAR RED INVALIDA
```

```
deny ip 0.0.0.0 0.255.255.255 any
```

```
remark BLOQUEAR TELNET
```

```
deny tcp any any eq 23
```

```
remark PERMITIR REDES PRIVADAS
```

```
permit ip 192.168.0.0 0.0.255.255 any
```

```
remark DENEGAR TODO
```

```
deny ip any any log
```

```
ip access-list standard ACL_VTY_ACCESS
```

```
permit 172.16.99.0 0.0.0.31
```

```
deny any log
```

! Aplicar ACL en interfaz WAN

interface Serial0/0/0

ip access-group ACL_R4 in

! ===== SSH SOLO EN VTY =====

line vty 0 4

transport input ssh

login local

access-class ACL_VTY_ACCESS in

exec-timeout 5 0

logging synchronous

line con 0

password cisco123

login

exec-timeout 10 0

logging synchronous

end

SW0 -----

enable

configure terminal

hostname SW0

no ip domain-lookup

banner motd # ADVERTENCIA: SW0 - Solo personal autorizado. #

username admin privilege 15 secret cisco123

enable secret cisco123

```
ip domain-name empresa.local
```

```
service password-encryption
```

```
crypto key generate rsa general-keys modulus 1024
```

```
ip ssh version 2
```

```
! ===== VLANS =====
```

```
vlan 10
```

```
name TI
```

```
vlan 20
```

```
name RRHH
```

```
vlan 99
```

```
name ADMIN
```

```
! ===== CAMBIAR VLAN NATIVA =====
```

```
vlan 999
```

```
name NATIVE
```

```
! ===== TRUNK UPLINK =====
```

```
interface f0/1
```

```
switchport trunk encapsulation dot1q
```

```
switchport mode trunk
```

```
switchport nonegotiate
```

```
switchport trunk native vlan 999
```

```
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100,110,120,99,999
```

```
no shutdown
```

```
interface f0/6
```

```
switchport trunk encapsulation dot1q
```

```
switchport mode trunk
```

```
switchport nonegotiate
```

```
switchport trunk native vlan 999
```

```
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100,110,120,99,999
```

```
no shutdown
```

```
! ===== ACCESS PORTS =====
```

```
interface FastEthernet0/2
```

```
switchport mode access
```

```
switchport access vlan 10
```

```
switchport nonegotiate
```

```
switchport port-security maximum 2
```

```
switchport port-security violation shutdown
```

```
switchport port-security
```

```
spanning-tree portfast
```

```
switchport port-security mac-address sticky
```

```
no shutdown
```

```
interface FastEthernet0/3
```

```
switchport mode access
```

```
switchport access vlan 10
```

```
switchport nonegotiate
```

```
switchport port-security maximum 2
```

```
switchport port-security violation shutdown
```

```
switchport port-security
```

```
switchport port-security mac-address sticky
```

```
spanning-tree portfast
```

```
no shutdown
```

```
interface FastEthernet0/4
```

```
switchport mode access
```

```
switchport access vlan 10
```

switchport nonegotiate

switchport port-security maximum 2

switchport port-security violation shutdown

switchport port-security

switchport port-security mac-address sticky

spanning-tree portfast

no shutdown

! ===== APAGAR INTERFACES NO USADAS =====

interface range FastEthernet0/7 - 24

shutdown

! ===== VLAN ADMIN E IP =====

interface vlan 99

ip address 172.16.99.2 255.255.255.224

no shutdown

ip default-gateway 172.16.99.1

=====ACL=====

ip access-list standard ACL_VTY

permit 172.16.99.0 0.0.0.31

deny any

line vty 0 4

access-class ACL_VTY in

transport input ssh

login local

exec-timeout 5 0

logging synchronous

line con 0

```
password cisco123
```

```
login
```

```
logging synchronous
```

```
end
```

```
-----  
SW1 -----  
-----
```

```
enable
```

```
configure terminal
```

```
hostname SW1
```

```
no ip domain-lookup
```

```
banner motd # ADVERTENCIA: SW1 - Solo personal autorizado. #
```

```
username admin privilege 15 secret cisco123
```

```
enable secret cisco123
```

```
service password-encryption
```

```
ip domain-name empresa.local
```

```
crypto key generate rsa general-keys modulus 1024
```

```
ip ssh version 2
```

```
! ===== VLANS =====
```

```
vlan 10
```

```
name TI
```

```
vlan 20
```

```
name RRHH
```

```
vlan 99
```

```
name ADMIN
```

```
! ===== CAMBIAR VLAN NATIVA =====
```

vlan 999

name NATIVE

! ===== TRUNK UPLINK =====

interface f0/2

switchport trunk encapsulation dot1q

switchport mode trunk

switchport nonegotiate

switchport trunk native vlan 999

switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100,110,120,99,999

no shutdown

! ===== ACCESS PORTS =====

interface FastEthernet0/3

switchport mode access

switchport access vlan 20

switchport nonegotiate

switchport port-security maximum 2

switchport port-security violation shutdown

switchport port-security

switchport port-security mac-address sticky

spanning-tree portfast

no shutdown

interface FastEthernet0/4

switchport mode access

switchport access vlan 20

switchport nonegotiate

switchport port-security maximum 2

switchport port-security violation shutdown

switchport port-security

switchport port-security mac-address sticky

spanning-tree portfast

no shutdown

interface FastEthernet0/5

switchport mode access

switchport access vlan 20

switchport nonegotiate

switchport port-security maximum 2

switchport port-security violation shutdown

switchport port-security

switchport port-security mac-address sticky

spanning-tree portfast

no shutdown

! ===== APAGAR INTERFACES NO USADAS =====

interface range FastEthernet0/8 - 24

shutdown

=====ACL=====

ip access-list standard ACL_VTY

permit 172.16.99.0 0.0.0.31

deny any

! ===== VLAN ADMIN E IP =====

interface vlan 99

ip address 172.16.99.3 255.255.255.224

no shutdown

ip default-gateway 172.16.99.1

line vty 0 4

```
access-class ACL_VTY in
transport input ssh
login local
exec-timeout 5 0
logging synchronous
```

```
line con 0
password cisco123
login
logging synchronous
```

```
=====
=====
```

Configuración de la provincia 2

```
Provincia 2-----
-----
```

```
R5(OSPF Area 0 - Router on Stick)
enable
configure terminal
hostname R5
no ip domain-lookup
banner motd # ADVERTENCIA: R5 - Acceso solo personal autorizado. #
```

```
username admin privilege 15 secret cisco123
enable secret cisco123
service password-encryption
```

```
ip domain-name empresa.local
```

crypto key generate rsa general-keys modulus 2048

ip ssh version 2

ip ssh time-out 60

ip ssh authentication-retries 3

login block-for 120 attempts 3 within 60

! ===== DHCP EXCLUDE =====

ip dhcp excluded-address 192.168.30.1

ip dhcp excluded-address 192.168.40.1

ip dhcp excluded-address 172.16.99.4

! ===== DHCP POOLS =====

ip dhcp pool VLAN30

network 192.168.30.0 255.255.255.0

default-router 192.168.30.1

dns-server 8.8.8.8

ip dhcp pool VLAN40

network 192.168.40.0 255.255.255.0

default-router 192.168.40.1

dns-server 8.8.8.8

! ===== INTERFACE WAN 1 =====

interface Serial0/1/0

description ENLACE WAN PRINCIPAL

ip address 10.0.0.26 255.255.255.252

no shutdown

! ===== INTERFACE WAN 2 =====

interface Serial0/2/0

description ENLACE WAN BACKUP

ip address 10.0.0.30 255.255.255.252

no shutdown

! ===== ROUTER ON STICK =====

```
interface GigabitEthernet0/1
description TRUNK AL SWITCH
no ip address
no shutdown
```

```
interface GigabitEthernet0/1.30
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
```

```
interface GigabitEthernet0/1.40
encapsulation dot1Q 40
ip address 192.168.40.1 255.255.255.0
```

! VLAN ADMINISTRATIVA=====0

```
interface GigabitEthernet0/1.99
encapsulation dot1Q 99
ip address 172.16.99.4 255.255.255.224
```

```
interface GigabitEthernet0/1.999
encapsulation dot1Q 999 native
no ip address
```

! ===== OSPF CON MD5 =====

```
router ospf 1
router-id 5.5.5.5
area 0 authentication message-digest
network 10.0.0.24 0.0.0.3 area 0
network 10.0.0.28 0.0.0.3 area 0
network 192.168.30.0 0.0.0.255 area 0
```

```
network 192.168.40.0 0.0.0.255 area 0
```

```
network 172.16.99.0 0.0.0.31 area 0
```

```
interface Serial0/1/0
```

```
ip ospf message-digest-key 1 md5 cisco123
```

```
ip ospf authentication message-digest
```

```
interface Serial0/2/0
```

```
ip ospf message-digest-key 1 md5 cisco123
```

```
ip ospf authentication message-digest
```

```
! ===== RUTA ESTATICA HACIA INTERNET =====
```

```
ip route 0.0.0.0 0.0.0.0 10.0.0.25
```

```
ip route 0.0.0.0 0.0.0.0 10.0.0.29 5
```

```
! ===== ACL DE SEGURIDAD =====
```

```
ip access-list extended ACL_WAN_IN
```

```
remark PERMITIR OSPF (CRUCIAL PARA LA ADYACENCIA)
```

```
permit ospf any any
```

```
remark BLOQUEAR TELNET (REQUISITO)
```

```
deny tcp any any eq 23
```

```
remark PERMITIR TRÁFICO DE REDES INTERNAS
```

```
permit ip 192.168.0.0 0.0.255.255 any
```

```
permit ip 172.16.99.0 0.0.0.31 any
```

```
remark DENEGAR EL RESTO Y LOGUEAR
```

```
deny ip any any log
```

```
! ===== APLICACIÓN CORRECTA EN WAN =====
```

```
interface Serial0/1/0
```

```
ip access-group ACL_WAN_IN in
```

```
interface Serial0/2/0
```

```
ip access-group ACL_WAN_IN in
```

```
! ACL : Control de VTY
```

```
ip access-list standard ACL_VTY_ACCESS
```

```
permit 172.16.99.0 0.0.0.31
```

```
deny any log
```

```
! ACL 3: Limitar ICMP (Seguridad Pertinente)
```

```
ip access-list extended ACL_PING_LIMIT
```

```
permit icmp any any echo-reply
```

```
permit icmp any any unreachable
```

```
deny icmp any any
```

```
! ACL 4: Bloquear tráfico basura (Anti-Spoofing)
```

```
ip access-list standard ACL_NO_SPOOF
```

```
deny 127.0.0.0 0.255.255.255
```

```
permit any
```

```
! ===== SSH SOLO EN VTY =====
```

```
line vty 0 4
```

```
transport input ssh
```

```
login local
```

```
access-class ACL_VTY_ACCESS in
```

```
exec-timeout 5 0
```

```
logging synchronous
```

```
line con 0
```

```
password cisco123
```

```
login
```

```
exec-timeout 10 0
```

```
logging synchronous
```

```
end
```

```
-----
```

```
SW2 -----
```

```
-----
```

```
enable
```

```
configure terminal
```

```
hostname SW2
```

```
no ip domain-lookup
```

```
banner motd # ADVERTENCIA: SW2 - Solo personal autorizado. #
```

```
username admin privilege 15 secret cisco123
```

```
enable secret cisco123
```

```
service password-encryption
```

```
ip domain-name empresa.local
```

```
crypto key generate rsa general-keys modulus 1024
```

```
ip ssh version 2
```

```
! ===== VLANS =====
```

```
vlan 30
```

```
name contabilidad
```

```
vlan 40
```

```
name marketing
```

```
vlan 99
```

```
name ADMIN
```

```
! ===== CAMBIAR VLAN NATIVA =====
```

```
vlan 999
```

```
name UNUSED_NATIVE
```

```
! ===== TRUNK UPLINK =====
```

```
interface f0/1
```

```
switchport mode trunk
```

```
switchport nonegotiate
```

```
switchport trunk native vlan 999
```

```
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100,110,120,99,999
```

```
no shutdown
```

```
interface g0/2
```

```
switchport mode trunk
```

```
switchport nonegotiate
```

```
switchport trunk native vlan 999
```

```
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100,110,120,99,999
```

```
no shutdown
```

```
! ===== ACCESS PORTS =====
```

```
interface FastEthernet0/2
```

```
switchport mode access
```

```
switchport access vlan 30
```

```
switchport nonegotiate
```

```
switchport port-security maximum 2
```

```
switchport port-security violation shutdown
```

```
switchport port-security
```

```
switchport port-security mac-address sticky
```

```
spanning-tree portfast
```

```
no shutdown
```

```
interface FastEthernet0/3
```

```
switchport mode access
```

```
switchport access vlan 30
switchport nonegotiate
switchport port-security maximum 2
switchport port-security violation shutdown
switchport port-security
switchport port-security mac-address sticky
spanning-tree portfast
no shutdown

! ===== APAGAR INTERFACES NO USADAS =====

interface range FastEthernet0/5 - 24
shutdown

! ===== VLAN ADMIN E IP =====

interface vlan 99
ip address 172.16.99.5 255.255.255.224
no shutdown

ip default-gateway 172.16.99.4

=====ACL=====

ip access-list standard ACL_VTY
permit 172.16.99.0 0.0.0.31
deny any

line vty 0 4
transport input ssh
login local
access-class ACL_VTY in
exec-timeout 5 0
logging synchronous
```

```
line con 0
password cisco123
login
logging synchronous
end
```

```
-----
SW3 (R5 - VLAN40)-----
-----
```

```
enable
configure terminal
hostname SW3
no ip domain-lookup
banner motd # ADVERTENCIA: SW3 - Solo personal autorizado. #
```

```
username admin privilege 15 secret cisco123
enable secret cisco123
service password-encryption
ip domain-name empresa.local
crypto key generate rsa general-keys modulus 1024
ip ssh version 2
```

```
! ===== VLANS =====
```

```
vlan 30
name contabilidad
```

```
vlan 40
name marketing
```

```
vlan 99
```

```
name ADMIN
```

```
! ===== CAMBIAR VLAN NATIVA =====
```

```
vlan 999
```

```
name UNUSED_NATIVE
```

```
! ===== TRUNK UPLINK =====
```

```
interface f0/2
```

```
switchport mode trunk
```

```
switchport nonegotiate
```

```
switchport trunk native vlan 999
```

```
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100,110,120,99,999
```

```
no shutdown
```

```
! ===== ACCESS PORTS =====
```

```
interface FastEthernet0/3
```

```
switchport mode access
```

```
switchport access vlan 40
```

```
switchport nonegotiate
```

```
switchport port-security maximum 2
```

```
switchport port-security violation shutdown
```

```
switchport port-security
```

```
switchport port-security mac-address sticky
```

```
spanning-tree portfast
```

```
no shutdown
```

```
interface FastEthernet0/4
```

```
switchport mode access
```

```
switchport access vlan 40
```

```
switchport nonegotiate
```

```
switchport port-security maximum 2
```

switchport port-security violation shutdown

switchport port-security

switchport port-security mac-address sticky

spanning-tree portfast

no shutdown

! ===== APAGAR INTERFACES NO USADAS =====

interface range FastEthernet0/6 - 24

shutdown

! ===== VLAN ADMIN E IP =====

interface vlan 99

ip address 172.16.99.6 255.255.255.224

no shutdown

ip default-gateway 172.16.99.4

=====ACL=====

ip access-list standard ACL_VTY

permit 172.16.99.0 0.0.0.31

deny any

line vty 0 4

transport input ssh

login local

access-class ACL_VTY in

exec-timeout 5 0

logging synchronous

line con 0

password cisco123

login

logging synchronous

end

Configuración de la provincia 3

provincia 3-----

R6-Provincia3 (OSPF Area 0 - Router on Stick)

enable

configure terminal

hostname R6

no ip domain-lookup

banner motd # ADVERTENCIA: R6-Provincia3 - Acceso solo personal autorizado. #

username admin privilege 15 secret cisco123

enable secret cisco123

service password-encryption

ip domain-name empresa.local

crypto key generate rsa general-keys modulus 1024

ip ssh version 2

ip ssh time-out 60

ip ssh authentication-retries 3

login block-for 120 attempts 3 within 60

! ===== DHCP EXCLUDE =====

ip dhcp excluded-address 192.168.50.1

ip dhcp excluded-address 192.168.60.1

ip dhcp excluded-address 192.168.70.1

ip dhcp excluded-address 192.168.80.1

ip dhcp excluded-address 172.16.99.7

! ===== DHCP POOLS =====

```
ip dhcp pool VLAN50
network 192.168.50.0 255.255.255.0
default-router 192.168.50.1
dns-server 8.8.8.8
```

```
ip dhcp pool VLAN60
network 192.168.60.0 255.255.255.0
default-router 192.168.60.1
dns-server 8.8.8.8
```

```
ip dhcp pool VLAN70
network 192.168.70.0 255.255.255.0
default-router 192.168.70.1
dns-server 8.8.8.8
```

```
ip dhcp pool VLAN80
network 192.168.80.0 255.255.255.0
default-router 192.168.80.1
dns-server 8.8.8.8
```

```
! ===== INTERFACE WAN 1 =====
```

```
interface Serial0/1/1
description ENLACE WAN PRINCIPAL
ip address 10.0.0.34 255.255.255.252
no shutdown
```

```
! ===== INTERFACE WAN 2 =====
```

```
interface Serial0/2/1
description ENLACE WAN BACKUP
ip address 10.0.0.38 255.255.255.252
no shutdown
```

! ===== ROUTER ON STICK =====

```
interface GigabitEthernet0/1
description TRUNK AL SWITCH
no ip address
no shutdown
```

```
interface GigabitEthernet0/1.50
encapsulation dot1Q 50
ip address 192.168.50.1 255.255.255.0
```

```
interface GigabitEthernet0/1.60
encapsulation dot1Q 60
ip address 192.168.60.1 255.255.255.0
```

```
interface GigabitEthernet0/1.70
encapsulation dot1Q 70
ip address 192.168.70.1 255.255.255.0
```

```
interface GigabitEthernet0/1.80
encapsulation dot1Q 80
ip address 192.168.80.1 255.255.255.0
```

! VLAN ADMINISTRATIVA=====0

```
interface GigabitEthernet0/1.99
encapsulation dot1Q 99
ip address 172.16.99.7 255.255.255.224
```

```
interface GigabitEthernet0/1.999
encapsulation dot1Q 999 native
```

no ip address

! ===== OSPF CON MD5 =====

router ospf 1

router-id 6.6.6.6

area 0 authentication message-digest

network 10.0.0.32 0.0.0.3 area 0

network 10.0.0.36 0.0.0.3 area 0

network 192.168.50.0 0.0.0.255 area 0

network 192.168.60.0 0.0.0.255 area 0

network 192.168.70.0 0.0.0.255 area 0

network 192.168.80.0 0.0.0.255 area 0

network 172.16.99.0 0.0.0.31 area 0

interface Serial0/1/1

ip ospf message-digest-key 1 md5 cisco123

ip ospf authentication message-digest

interface Serial0/2/1

ip ospf message-digest-key 1 md5 cisco123

ip ospf authentication message-digest

! ===== RUTA ESTATICA HACIA INTERNET =====

ip route 0.0.0.0 0.0.0.0 10.0.0.33

ip route 0.0.0.0 0.0.0.0 10.0.0.37 5

! ===== ACL DE SEGURIDAD =====

ip access-list extended ACL_WAN_IN

remark PERMITIR OSPF (CRUCIAL PARA LA ADYACENCIA)

permit ospf any any

```
remark BLOQUEAR TELNET (REQUISITO)
```

```
deny tcp any any eq 23
```

```
remark PERMITIR TRÁFICO DE REDES INTERNAS
```

```
permit ip 192.168.0.0 0.0.255.255 any
```

```
permit ip 172.16.99.0 0.0.0.31 any
```

```
remark DENEGAR EL RESTO Y LOGUEAR
```

```
deny ip any any log
```

```
! ===== APLICACIÓN CORRECTA EN WAN =====
```

```
interface Serial0/1/1
```

```
ip access-group ACL_WAN_IN in
```

```
interface Serial0/2/1
```

```
ip access-group ACL_WAN_IN in
```

```
! ACL: Control de VTY=====
```

```
ip access-list standard ACL_VTY_ACCESS
```

```
permit 172.16.99.0 0.0.0.31
```

```
deny any log
```

```
! ===== SSH SOLO EN VTY =====
```

```
line vty 0 4
```

```
transport input ssh
```

```
login local
```

```
access-class ACL_VTY_ACCESS in
```

```
exec-timeout 5 0
```

```
logging synchronous
```

```
line con 0
```

```
password cisco123
```

```
login
```

```
exec-timeout 10 0
```

```
logging synchronous
```

```
end
```

```
-----
```

```
SW4 -----
```

```
-----
```

```
enable
```

```
configure terminal
```

```
hostname SW4
```

```
no ip domain-lookup
```

```
banner motd # ADVERTENCIA: SW4 (R6 - VLAN50/60) - Solo personal autorizado. #
```

```
username admin privilege 15 secret cisco123
```

```
enable secret cisco123
```

```
service password-encryption
```

```
ip domain-name empresa.local
```

```
crypto key generate rsa general-keys modulus 1024
```

```
ip ssh version 2
```

```
! ===== VLANS =====
```

```
vlan 50
```

```
name ventas
```

```
vlan 60
```

```
name finanzas
```

```
vlan 70
```

name logistica

vlan 80

name calidad

vlan 99

name ADMIN

! ===== CAMBIAR VLAN NATIVA =====

vlan 999

name UNUSED_NATIVE

! ===== TRUNK UPLINK =====

interface f0/1

switchport mode trunk

switchport nonegotiate

switchport trunk native vlan 999

switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100,110,120,99,999

no shutdown

! ===== ACCESS PORTS =====

interface FastEthernet0/2

switchport mode access

switchport access vlan 50

switchport nonegotiate

switchport port-security maximum 2

switchport port-security violation shutdown

switchport port-security

switchport port-security mac-address sticky

spanning-tree portfast

no shutdown

```
interface FastEthernet0/3
  switchport mode access
  switchport access vlan 60
  switchport nonegotiate
  switchport port-security maximum 2
  switchport port-security violation shutdown
  switchport port-security
  switchport port-security mac-address sticky
  spanning-tree portfast
  no shutdown

! ===== APAGAR INTERFACES NO USADAS =====

interface range FastEthernet0/5 - 24
  shutdown

! ===== VLAN ADMIN E IP =====

interface vlan 99
  ip address 172.16.99.8 255.255.255.224
  no shutdown

ip default-gateway 172.16.99.7

=====ACL=====

ip access-list standard ACL_VTY
  permit 172.16.99.0 0.0.0.31
  deny any

line vty 0 4
  transport input ssh
  login local
  access-class ACL_VTY in
  exec-timeout 5 0
```

logging synchronous

line con 0

password cisco123

login

logging synchronous

end

SW5 (R6 - VLAN70/80)-----

enable

configure terminal

hostname SW5

no ip domain-lookup

banner motd # ADVERTENCIA: SW5 (R6 - VLAN70/80) - Solo personal autorizado. #

username admin privilege 15 secret cisco123

enable secret cisco123

service password-encryption

ip domain-name empresa.local

crypto key generate rsa general-keys modulus 1024

ip ssh version 2

! ===== VLANS =====

vlan 50

name ventas

vlan 60

name finanzas

vlan 70

name logistica

vlan 80

name calidad

vlan 99

name ADMIN

! ===== CAMBIAR VLAN NATIVA =====

vlan 999

name UNUSED_NATIVE

! ===== TRUNK UPLINK =====

interface f0/3

switchport mode trunk

switchport nonegotiate

switchport trunk native vlan 999

switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100,110,120,99,999

no shutdown

! ===== ACCESS PORTS =====

interface FastEthernet0/1

switchport mode access

switchport access vlan 70

switchport nonegotiate

switchport port-security maximum 2

switchport port-security violation shutdown

switchport port-security

switchport port-security mac-address sticky

spanning-tree portfast

no shutdown

interface FastEthernet0/2

switchport mode access

switchport access vlan 80

switchport nonegotiate

switchport port-security maximum 2

switchport port-security violation shutdown

switchport port-security

switchport port-security mac-address sticky

spanning-tree portfast

no shutdown

! ===== APAGAR INTERFACES NO USADAS =====

interface range FastEthernet0/5 - 24

shutdown

! ===== VLAN ADMIN E IP =====

interface vlan 99

ip address 172.16.99.9 255.255.255.224

no shutdown

ip default-gateway 172.16.99.7

=====ACL=====

ip access-list standard ACL_VTY

permit 172.16.99.0 0.0.0.31

deny any

line vty 0 4

transport input ssh

```
login local
access-class ACL_VTY in
exec-timeout 5 0
logging synchronous
```

```
line con 0
password cisco123
login
logging synchronous
end
```

```
-----
SW6 (R6 - VLAN70/80,50,60)-----
-----
```

```
enable
configure terminal
hostname SW6
no ip domain-lookup
banner motd # ADVERTENCIA: SW6 - Solo personal autorizado. #
```

```
username admin privilege 15 secret cisco123
enable secret cisco123
service password-encryption
ip domain-name empresa.local
crypto key generate rsa general-keys modulus 1024
ip ssh version 2
! ===== VLANS =====
```

```
vlan 50
```

name ventas

vlan 60

name finanzas

vlan 70

name logistica

vlan 80

name calidad

vlan 99

name ADMIN

! ===== CAMBIAR VLAN NATIVA =====

vlan 999

name UNUSED_NATIVE

! ===== TRUNK UPLINK =====

interface f0/1

switchport mode trunk

switchport nonegotiate

switchport trunk native vlan 999

switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100,110,120,99,999

no shutdown

interface f0/2

switchport mode trunk

switchport nonegotiate

switchport trunk native vlan 999

switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100,110,120,99,999

no shutdown

interface f0/3

switchport mode trunk

switchport nonegotiate

switchport trunk native vlan 999

switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100,110,120,99,999

no shutdown

! ===== APAGAR INTERFACES NO USADAS =====

interface range FastEthernet0/5 - 24

shutdown

! ===== VLAN ADMIN E IP =====

interface vlan 99

ip address 172.16.99.10 255.255.255.224

no shutdown

ip default-gateway 172.16.99.7

=====ACL=====

ip access-list standard ACL_VTY

permit 172.16.99.0 0.0.0.31

deny any

line vty 0 4

transport input ssh

login local

access-class ACL_VTY in

exec-timeout 5 0

logging synchronous

```
line con 0
password cisco123
login
logging synchronous
end
```

Configuración de la provincia 4

```
Provincia 4-----
-----

R7(OSPF Area 0 - Router on Stick)

enable
configure terminal
hostname R7
no ip domain-lookup
banner motd # ADVERTENCIA: R7 - Acceso solo personal autorizado. #

username admin privilege 15 secret cisco123
enable secret cisco123
service password-encryption

ip domain-name empresa.local
crypto key generate rsa general-keys modulus 2048
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3
login block-for 120 attempts 3 within 60
! ===== DHCP EXCLUDE =====
```

ip dhcp excluded-address 192.168.90.1

ip dhcp excluded-address 192.168.100.1

ip dhcp excluded-address 172.16.99.11

! ===== DHCP POOLS =====

ip dhcp pool VLAN90

network 192.168.90.0 255.255.255.0

default-router 192.168.90.1

dns-server 8.8.8.8

ip dhcp pool VLAN100

network 192.168.100.0 255.255.255.0

default-router 192.168.100.1

dns-server 8.8.8.8

! ===== INTERFACE WAN 1 =====

interface Serial0/2/0

description ENLACE WAN PRINCIPAL

ip address 10.0.0.42 255.255.255.252

no shutdown

! ===== INTERFACE WAN 2 =====

interface Serial0/0/1

description ENLACE WAN BACKUP

ip address 10.0.0.46 255.255.255.252

no shutdown

! ===== ROUTER ON STICK =====

interface GigabitEthernet0/1

description TRUNK AL SWITCH

no ip address

no shutdown

```
interface GigabitEthernet0/1.90
 encapsulation dot1Q 90
 ip address 192.168.90.1 255.255.255.0
```

```
interface GigabitEthernet0/1.100
 encapsulation dot1Q 100
 ip address 192.168.100.1 255.255.255.0
```

```
! VLAN ADMINISTRATIVA=====
```

```
interface GigabitEthernet0/1.99
 encapsulation dot1Q 99
 ip address 172.16.99.11 255.255.255.224
```

```
interface GigabitEthernet0/1.999
 encapsulation dot1Q 999 native
 no ip address
```

```
! ===== OSPF CON MD5 =====
```

```
router ospf 1
 router-id 7.7.7.7
 area 0 authentication message-digest
 network 10.0.0.40 0.0.0.3 area 0
 network 10.0.0.44 0.0.0.3 area 0
 network 192.168.90.0 0.0.0.255 area 0
 network 192.168.100.0 0.0.0.255 area 0
 network 172.16.99.0 0.0.0.31 area 0
```

```
interface Serial0/0/1
 ip ospf message-digest-key 1 md5 cisco123
```

```
ip ospf authentication message-digest
```

```
interface Serial0/2/0
```

```
ip ospf message-digest-key 1 md5 cisco123
```

```
ip ospf authentication message-digest
```

```
! ===== RUTA ESTATICA HACIA INTERNET =====
```

```
ip route 0.0.0.0 0.0.0.0 10.0.0.41
```

```
ip route 0.0.0.0 0.0.0.0 10.0.0.45 5
```

```
! ===== ACL DE SEGURIDAD =====
```

```
ip access-list extended ACL_WAN_IN
```

```
remark PERMITIR OSPF (CRUCIAL PARA LA ADYACENCIA)
```

```
permit ospf any any
```

```
remark BLOQUEAR TELNET (REQUISITO)
```

```
deny tcp any any eq 23
```

```
remark PERMITIR TRÁFICO DE REDES INTERNAS
```

```
permit ip 192.168.0.0 0.0.255.255 any
```

```
permit ip 172.16.99.0 0.0.0.31 any
```

```
remark DENEGAR EL RESTO Y LOGUEAR
```

```
deny ip any any log
```

```
! ===== APLICACIÓN CORRECTA EN WAN =====
```

```
interface Serial0/0/1
```

```
ip access-group ACL_WAN_IN in
```

```
interface Serial0/2/0
```

```
ip access-group ACL_WAN_IN in
```

```
! ACL : Control de VTY
```

```
ip access-list standard ACL_VTY_ACCESS
permit 172.16.99.0 0.0.0.31
deny any log
```

! ACL 3: Limitar ICMP (Seguridad Pertinente)

```
ip access-list extended ACL_PING_LIMIT
permit icmp any any echo-reply
permit icmp any any unreachable
deny icmp any any
```

! ACL 4: Bloquear tráfico basura (Anti-Spoofing)

```
ip access-list standard ACL_NO_SPOOF
deny 127.0.0.0 0.255.255.255
permit any
```

! ===== SSH SOLO EN VTY =====

```
line vty 0 4
transport input ssh
login local
access-class ACL_VTY_ACCESS in
exec-timeout 5 0
logging synchronous
```

```
line con 0
password cisco123
login
exec-timeout 10 0
logging synchronous
end
```

SW7-r7 -----

enable

configure terminal

hostname SW7-r7

no ip domain-lookup

banner motd # ADVERTENCIA: SW7 - Solo personal autorizado. #

username admin privilege 15 secret cisco123

enable secret cisco123

service password-encryption

ip domain-name empresa.local

crypto key generate rsa general-keys modulus 1024

ip ssh version 2

! ===== VLANS =====

vlan 90

name compras

vlan 100

name almacén

vlan 99

name ADMIN

! ===== CAMBIAR VLAN NATIVA =====

vlan 999

name UNUSED_NATIVE

! ===== TRUNK UPLINK =====

interface g0/1

switchport mode trunk

```
switchport nonegotiate
switchport trunk native vlan 999
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100,110,120,99,999
no shutdown
```

```
interface g0/2
switchport mode trunk
switchport nonegotiate
switchport trunk native vlan 999
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100,110,120,99,999
no shutdown
```

! ===== ACCESS PORTS =====

```
interface FastEthernet0/1
switchport mode access
switchport access vlan 90
switchport nonegotiate
switchport port-security maximum 2
switchport port-security violation shutdown
switchport port-security
switchport port-security mac-address sticky
spanning-tree portfast
no shutdown
```

```
interface FastEthernet0/2
switchport mode access
switchport access vlan 90
switchport nonegotiate
switchport port-security maximum 2
switchport port-security violation shutdown
```

```
switchport port-security
switchport port-security mac-address sticky
spanning-tree portfast
no shutdown
```

```
interface FastEthernet0/3
switchport mode access
switchport access vlan 90
switchport nonegotiate
switchport port-security maximum 2
switchport port-security violation shutdown
switchport port-security
switchport port-security mac-address sticky
spanning-tree portfast
no shutdown
```

! ===== APAGAR INTERFACES NO USADAS =====

```
interface range FastEthernet0/5 - 24
shutdown
```

! ===== VLAN ADMIN E IP =====

```
interface vlan 99
ip address 172.16.99.12 255.255.255.224
no shutdown
```

```
ip default-gateway 172.16.99.11
```

=====ACL=====

```
ip access-list standard ACL_VTY
permit 172.16.99.0 0.0.0.31
```

deny any

line vty 0 4

transport input ssh

login local

access-class ACL_VTY in

exec-timeout 5 0

logging synchronous

line con 0

password cisco123

login

logging synchronous

end

SW8 (R7 - VLAN100)-----

enable

configure terminal

hostname SW8-r7

no ip domain-lookup

banner motd # ADVERTENCIA: SW8 - Solo personal autorizado. #

username admin privilege 15 secret cisco123

enable secret cisco123

service password-encryption

ip domain-name empresa.local

crypto key generate rsa general-keys modulus 1024

```
ip ssh version 2
```

```
! ===== VLANS =====
```

```
vlan 90
```

```
name compras
```

```
vlan 100
```

```
name almacen
```

```
vlan 99
```

```
name ADMIN
```

```
! ===== CAMBIAR VLAN NATIVA =====
```

```
vlan 999
```

```
name UNUSED_NATIVE
```

```
! ===== TRUNK UPLINK =====
```

```
interface g0/2
```

```
switchport mode trunk
```

```
switchport nonegotiate
```

```
switchport trunk native vlan 999
```

```
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100,110,120,99,999
```

```
no shutdown
```

```
! ===== ACCESS PORTS =====
```

```
interface FastEthernet0/1
```

```
switchport mode access
```

```
switchport access vlan 100
```

```
switchport nonegotiate
```

```
switchport port-security maximum 2
```

```
switchport port-security violation shutdown
```

```
switchport port-security
```

```
switchport port-security mac-address sticky
```

```
spanning-tree portfast
```

no shutdown

interface FastEthernet0/2

switchport mode access

switchport access vlan 100

switchport nonegotiate

switchport port-security maximum 2

switchport port-security violation shutdown

switchport port-security

switchport port-security mac-address sticky

spanning-tree portfast

no shutdown

interface FastEthernet0/3

switchport mode access

switchport access vlan 100

switchport nonegotiate

switchport port-security maximum 2

switchport port-security violation shutdown

switchport port-security

switchport port-security mac-address sticky

spanning-tree portfast

no shutdown

! ===== APAGAR INTERFACES NO USADAS =====

interface range FastEthernet0/6 - 24

shutdown

! ===== VLAN ADMIN E IP =====

interface vlan 99

ip address 172.16.99.13 255.255.255.224

no shutdown

ip default-gateway 172.16.99.11

=====ACL=====

ip access-list standard ACL_VTY

permit 172.16.99.0 0.0.0.31

deny any

line vty 0 4

transport input ssh

login local

access-class ACL_VTY in

exec-timeout 5 0

logging synchronous

line con 0

password cisco123

login

logging synchronous

end

=====0

Configuración de la provincia 5

Provincia 5-----

R8 (OSPF Area 0 - Router on Stick)

enable

configure terminal

```
hostname R8
no ip domain-lookup
banner motd # ADVERTENCIA: R8 - Acceso solo personal autorizado. #
```

```
username admin privilege 15 secret cisco123
enable secret cisco123
service password-encryption
ip domain-name empresa.local
```

```
crypto key generate rsa general-keys modulus 2048
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3
login block-for 120 attempts 3 within 60
```

```
! ===== DHCP EXCLUDE =====
ip dhcp excluded-address 192.168.110.1
ip dhcp excluded-address 192.168.120.1
ip dhcp excluded-address 172.16.99.14
```

```
! ===== DHCP POOLS =====
```

```
ip dhcp pool VLAN110
network 192.168.110.0 255.255.255.0
default-router 192.168.110.1
dns-server 8.8.8.8
```

```
ip dhcp pool VLAN120
network 192.168.120.0 255.255.255.0
```

default-router 192.168.120.1

dns-server 8.8.8.8

! ===== INTERFACE WAN 1 =====

interface Serial0/2/1

description ENLACE WAN PRINCIPAL

ip address 10.0.0.50 255.255.255.252

no shutdown

! ===== INTERFACE WAN 2 =====

interface Serial0/0/0

description ENLACE WAN BACKUP

ip address 10.0.0.54 255.255.255.252

no shutdown

! ===== ROUTER ON STICK =====

interface GigabitEthernet0/1

description TRUNK AL SWITCH

no ip address

no shutdown

interface GigabitEthernet0/1.110

encapsulation dot1Q 110

ip address 192.168.110.1 255.255.255.0

interface GigabitEthernet0/1.120

encapsulation dot1Q 120

```
ip address 192.168.120.1 255.255.255.0
```

```
! VLAN ADMINISTRATIVA
```

```
interface GigabitEthernet0/1.99
```

```
encapsulation dot1Q 99
```

```
ip address 172.16.99.14 255.255.255.224
```

```
interface GigabitEthernet0/1.999
```

```
encapsulation dot1Q 999 native
```

```
no ip address
```

```
! ===== OSPF CON MD5 =====
```

```
router ospf 1
```

```
router-id 8.8.8.8
```

```
area 0 authentication message-digest
```

```
network 10.0.0.48 0.0.0.3 area 0
```

```
network 10.0.0.52 0.0.0.3 area 0
```

```
network 192.168.110.0 0.0.0.255 area 0
```

```
network 192.168.120.0 0.0.0.255 area 0
```

```
network 172.16.99.0 0.0.0.31 area 0
```

```
interface Serial0/2/1
```

```
ip ospf message-digest-key 1 md5 cisco123
```

```
ip ospf authentication message-digest
```

```
interface Serial0/0/0
```

```
ip ospf message-digest-key 1 md5 cisco123
```

```
ip ospf authentication message-digest
```

```
! ===== RUTA ESTATICA HACIA INTERNET =====
```

```
ip route 0.0.0.0 0.0.0.0 10.0.0.49
```

```
ip route 0.0.0.0 0.0.0.0 10.0.0.53
```

```
! ===== ACL DE SEGURIDAD =====
```

```
ip access-list extended ACL_WAN_IN
```

```
remark PERMITIR OSPF (CRUCIAL PARA LA ADYACENCIA)
```

```
permit ospf any any
```

```
remark BLOQUEAR TELNET (REQUISITO)
```

```
deny tcp any any eq 23
```

```
remark PERMITIR TRÁFICO DE REDES INTERNAS
```

```
permit ip 192.168.0.0 0.0.255.255 any
```

```
permit ip 172.16.99.0 0.0.0.31 any
```

```
remark DENEGAR EL RESTO Y LOGUEAR
```

```
deny ip any any log
```

```
! ===== APLICACIÓN CORRECTA EN WAN =====
```

```
interface Serial0/2/1
```

```
ip access-group ACL_WAN_IN in
```

```
interface Serial0/0/0
```

```
ip access-group ACL_WAN_IN in
```

! ACL: Control de VTY

ip access-list standard ACL_VTY_ACCESS

permit 172.16.99.0 0.0.0.31

deny any log

! ===== SSH SOLO EN VTY =====

line vty 0 4

transport input ssh

login local

access-class ACL_VTY_ACCESS in

exec-timeout 5 0

logging synchronous

line con 0

password cisco123

login

exec-timeout 10 0

logging synchronous

end

SW9 vlan110 -----

enable

configure terminal

hostname SW9

no ip domain-lookup

```
banner motd # ADVERTENCIA: SW9 (R8 - VLAN110) - Solo personal autorizado. #
```

```
username admin privilege 15 secret cisco123
```

```
enable secret cisco123
```

```
service password-encryption
```

```
ip domain-name empresa.local
```

```
crypto key generate rsa general-keys modulus 1024
```

```
ip ssh version 2
```

```
! ===== VLANS =====
```

```
vlan 110
```

```
name asistente
```

```
vlan 120
```

```
name CRS
```

```
vlan 99
```

```
name ADMIN
```

```
! ===== CAMBIAR VLAN NATIVA =====
```

```
vlan 999
```

```
name UNUSED_NATIVE
```

```
! ===== TRUNK UPLINK =====
```

```
interface f0/2
```

```
switchport mode trunk
```

```
switchport nonegotiate
```

```
switchport trunk native vlan 999
```

```
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100,110,120,99,999  
no shutdown
```

```
! ===== ACCESS PORTS =====
```

```
interface FastEthernet0/3  
switchport mode access  
switchport access vlan 110  
switchport nonegotiate  
switchport port-security maximum 2  
switchport port-security violation shutdown  
switchport port-security  
switchport port-security mac-address sticky  
spanning-tree portfast  
no shutdown
```

```
interface FastEthernet0/4  
switchport mode access  
switchport access vlan 110  
switchport nonegotiate  
switchport port-security maximum 2  
switchport port-security violation shutdown  
switchport port-security  
switchport port-security mac-address sticky  
spanning-tree portfast  
no shutdown
```

```
interface FastEthernet0/5  
switchport mode access  
switchport access vlan 110
```

```
switchport nonegotiate
switchport port-security maximum 2
switchport port-security violation shutdown
switchport port-security
switchport port-security mac-address sticky
spanning-tree portfast
no shutdown
```

! ===== APAGAR INTERFACES NO USADAS =====

```
interface range FastEthernet0/6 - 24
shutdown
```

! ===== VLAN ADMIN E IP =====

```
interface vlan 99
ip address 172.16.99.16 255.255.255.224
no shutdown
```

```
ip default-gateway 172.16.99.14
```

=====ACL=====

```
ip access-list standard ACL_VTY
permit 172.16.99.0 0.0.0.31
deny any
```

```
line vty 0 4
transport input ssh
login local
access-class ACL_VTY in
exec-timeout 5 0
```

logging synchronous

line con 0

password cisco123

login

logging synchronous

end

SW11 (R8 - VLAN120)-----

enable

configure terminal

hostname SW11

no ip domain-lookup

banner motd # ADVERTENCIA: SW11 (R8 - VLAN120) - Solo personal autorizado. #

username admin privilege 15 secret cisco123

enable secret cisco123

service password-encryption

ip domain-name empresa.local

crypto key generate rsa general-keys modulus 1024

ip ssh version 2

! ===== VLANS =====

vlan 110

name asistente

vlan 120

name CRS

vlan 99

name ADMIN

! ===== CAMBIAR VLAN NATIVA =====

vlan 999

name UNUSED_NATIVE

! ===== TRUNK UPLINK =====

interface f0/3

switchport mode trunk

switchport nonegotiate

switchport trunk native vlan 999

switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100,110,120,99,999

no shutdown

! ===== ACCESS PORTS =====

interface FastEthernet0/4

switchport mode access

switchport access vlan 120

switchport nonegotiate

switchport port-security maximum 2

switchport port-security violation shutdown

switchport port-security

switchport port-security mac-address sticky

spanning-tree portfast

no shutdown

```
interface FastEthernet0/5
switchport mode access
switchport access vlan 120
switchport nonegotiate
switchport port-security maximum 2
switchport port-security violation shutdown
switchport port-security
switchport port-security mac-address sticky
spanning-tree portfast
no shutdown
```

```
interface FastEthernet0/6
switchport mode access
switchport access vlan 120
switchport nonegotiate
switchport port-security maximum 2
switchport port-security violation shutdown
switchport port-security
switchport port-security mac-address sticky
spanning-tree portfast
no shutdown
```

! ===== APAGAR INTERFACES NO USADAS =====

```
interface range FastEthernet0/7 - 24
shutdown
```

! ===== VLAN ADMIN E IP =====

```
interface vlan 99
```

```
ip address 172.16.99.17 255.255.255.224
```

```
no shutdown
```

```
ip default-gateway 172.16.99.14
```

```
=====ACL=====
```

```
ip access-list standard ACL_VTY
```

```
permit 172.16.99.0 0.0.0.31
```

```
deny any
```

```
line vty 0 4
```

```
transport input ssh
```

```
login local
```

```
access-class ACL_VTY in
```

```
exec-timeout 5 0
```

```
logging synchronous
```

```
line con 0
```

```
password cisco123
```

```
login
```

```
logging synchronous
```

```
end
```

```
-----  
SW10 (R8 - VLAN110/120)-----  
-----
```

```
enable
```

```
configure terminal
```

```
hostname SW10  
no ip domain-lookup  
banner motd # ADVERTENCIA: SW10 - Solo personal autorizado. #
```

```
username admin privilege 15 secret cisco123  
enable secret cisco123  
service password-encryption  
ip domain-name empresa.local  
crypto key generate rsa general-keys modulus 1024  
ip ssh version 2
```

```
! ===== VLANS =====
```

```
vlan 110  
name asistente
```

```
vlan 120  
name CRS
```

```
vlan 99  
name ADMIN
```

```
! ===== CAMBIAR VLAN NATIVA =====
```

```
vlan 999  
name UNUSED_NATIVE
```

```
! ===== TRUNK UPLINK =====
```

```
interface f0/1  
switchport mode trunk  
switchport nonegotiate
```

```
switchport trunk native vlan 999
```

```
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100,110,120,99,999
```

```
no shutdown
```

```
interface f0/2
```

```
switchport mode trunk
```

```
switchport nonegotiate
```

```
switchport trunk native vlan 999
```

```
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100,110,120,99,999
```

```
no shutdown
```

```
interface f0/3
```

```
switchport mode trunk
```

```
switchport nonegotiate
```

```
switchport trunk native vlan 999
```

```
switchport trunk allowed vlan 10,20,30,40,50,60,70,80,90,100,110,120,99,999
```

```
no shutdown
```

```
! ===== APAGAR INTERFACES NO USADAS =====
```

```
interface range FastEthernet0/5 - 24
```

```
shutdown
```

```
! ===== VLAN ADMIN E IP =====
```

```
interface vlan 99
```

```
ip address 172.16.99.15 255.255.255.224
```

```
no shutdown
```

```
ip default-gateway 172.16.99.14
```

=====ACL=====

```
ip access-list standard ACL_VTY
```

```
permit 172.16.99.0 0.0.0.31
```

```
deny any
```

```
line vty 0 4
```

```
transport input ssh
```

```
login local
```

```
access-class ACL_VTY in
```

```
exec-timeout 5 0
```

```
logging synchronous
```

```
line con 0
```

```
password cisco123
```

```
login
```

```
logging synchronous
```

```
end
```

Configuración de la Core R2

Core 2-----

R2-----

R2 (Core - Área 0)

```
enable
```

```
configure terminal
```

```
hostname R2
```

no ip domain-lookup

banner motd # ADVERTENCIA: Acceso restringido. Solo personal autorizado. #

username admin privilege 15 secret cisco123

enable secret cisco123

service password-encryption

ip domain-name empresa.local

crypto key generate rsa general-keys modulus 2048

ip ssh version 2

login block-for 120 attempts 3 within 60

! ===== INTERFACE WAN =====

interface Serial0/0/1

description R1-R2

ip address 10.0.0.6 255.255.255.252

no shutdown

interface Serial0/3/1

description R2-R3

ip address 10.0.0.13 255.255.255.252

no shutdown

interface Serial0/0/0

description R2-R4

ip address 10.0.0.17 255.255.255.252

no shutdown

interface Serial0/1/0

```
description R2-R5
ip address 10.0.0.25 255.255.255.252
no shutdown
```

```
interface Serial0/1/1
description R2-R6
ip address 10.0.0.33 255.255.255.252
no shutdown
```

```
interface Serial0/2/0
description R2-R7
ip address 10.0.0.41 255.255.255.252
no shutdown
```

```
interface Serial0/2/1
description R2-R8
ip address 10.0.0.49 255.255.255.252
no shutdown
```

```
router ospf 1
router-id 2.2.2.2
area 0 authentication message-digest
network 10.0.0.4 0.0.0.3 area 0
network 10.0.0.12 0.0.0.3 area 0
network 10.0.0.16 0.0.0.3 area 0
network 10.0.0.24 0.0.0.3 area 0
network 10.0.0.32 0.0.0.3 area 0
network 10.0.0.40 0.0.0.3 area 0
network 10.0.0.48 0.0.0.3 area 0
```

```
interface Serial0/3/1
ip ospf message-digest-key 1 md5 cisco123
ip ospf authentication message-digest
```

```
interface Serial0/2/0
ip ospf message-digest-key 1 md5 cisco123
ip ospf authentication message-digest
```

```
interface Serial0/2/1
ip ospf message-digest-key 1 md5 cisco123
ip ospf authentication message-digest
```

```
interface Serial0/0/0
ip ospf message-digest-key 1 md5 cisco123
ip ospf authentication message-digest
```

```
interface Serial0/1/0
ip ospf message-digest-key 1 md5 cisco123
ip ospf authentication message-digest
```

```
interface Serial0/1/1
ip ospf message-digest-key 1 md5 cisco123
ip ospf authentication message-digest
```

```
line vty 0 4
transport input ssh
login local
exec-timeout 5 0
```

logging synchronous

line con 0

password cisco123

login

logging synchronous

! ===== ACL DE SEGURIDAD =====

ip access-list standard ACL_VTY_ACCESS

permit 172.16.99.0 0.0.0.31

deny any

line vty 0 4

access-class ACL_VTY_ACCESS in

end

Configuración de la CORE R3

R3 (Core - Área 0)

enable

configure terminal

hostname R3

no ip domain-lookup

banner motd # ADVERTENCIA: Acceso restringido. Solo personal autorizado. #

username admin privilege 15 secret cisco123

enable secret cisco123

service password-encryption

ip domain-name empresa.local

crypto key generate rsa general-keys modulus 2048

ip ssh version 2

login block-for 120 attempts 3 within 60

! ===== INTERFACE WAN =====

interface Serial0/1/0

description R1-R3

ip address 10.0.0.10 255.255.255.252

no shutdown

interface Serial0/3/1

description R2-R3

ip address 10.0.0.14 255.255.255.252

no shutdown

interface Serial0/1/1

description R3-R4

ip address 10.0.0.21 255.255.255.252

no shutdown

interface Serial0/2/0

description R3-R5

ip address 10.0.0.29 255.255.255.252

no shutdown

interface Serial0/2/1

description R3-R6

ip address 10.0.0.37 255.255.255.252

no shutdown

interface Serial0/0/1

description R3-R7

ip address 10.0.0.45 255.255.255.252

no shutdown

interface Serial0/0/0

description R3-R8

ip address 10.0.0.53 255.255.255.252

no shutdown

router ospf 1

router-id 3.3.3.3

area 0 authentication message-digest

network 10.0.0.8 0.0.0.3 area 0

network 10.0.0.12 0.0.0.3 area 0

network 10.0.0.20 0.0.0.3 area 0

network 10.0.0.28 0.0.0.3 area 0

network 10.0.0.36 0.0.0.3 area 0

network 10.0.0.44 0.0.0.3 area 0

network 10.0.0.52 0.0.0.3 area 0

interface Serial0/1/0

ip ospf message-digest-key 1 md5 cisco123

ip ospf authentication message-digest

interface Serial0/3/1

ip ospf message-digest-key 1 md5 cisco123

```
ip ospf authentication message-digest
```

```
interface Serial0/1/1
```

```
ip ospf message-digest-key 1 md5 cisco123
```

```
ip ospf authentication message-digest
```

```
interface Serial0/2/0
```

```
ip ospf message-digest-key 1 md5 cisco123
```

```
ip ospf authentication message-digest
```

```
interface Serial0/2/1
```

```
ip ospf message-digest-key 1 md5 cisco123
```

```
ip ospf authentication message-digest
```

```
interface Serial0/0/1
```

```
ip ospf message-digest-key 1 md5 cisco123
```

```
ip ospf authentication message-digest
```

```
interface Serial0/0/0
```

```
ip ospf message-digest-key 1 md5 cisco123
```

```
ip ospf authentication message-digest
```

```
line vty 0 4
```

```
transport input ssh
```

```
login local
```

```
exec-timeout 5 0
```

```
logging synchronous
```

```
line con 0
```

```
password cisco123

login

logging synchronous

! ===== ACL DE SEGURIDAD =====

ip access-list standard ACL_VTY_ACCESS

permit 172.16.99.0 0.0.0.31

deny any


line vty 0 4

access-class ACL_VTY_ACCESS in


end
```

Configuración de la Core R1

```
R1 (Core - Área 0)

enable

configure terminal

hostname R1

no ip domain-lookup

banner motd # ADVERTENCIA: Acceso restringido. Solo personal autorizado. #


username admin privilege 15 secret cisco123

enable secret cisco123

service password-encryption


ip domain-name empresa.local

crypto key generate rsa general-keys modulus 2048

ip ssh version 2
```

login block-for 120 attempts 3 within 60

! ===== INTERFACES WAN =====

interface Serial0/0/0

description ISP-R1

ip address 210.10.10.2 255.255.255.252

no shutdown

interface Serial0/0/1

description R1-R2

ip address 10.0.0.5 255.255.255.252

no shutdown

interface Serial0/1/0

description R1-R3

ip address 10.0.0.9 255.255.255.252

no shutdown

! ===== OSPF CON MD5 =====

router ospf 1

router-id 1.1.1.1

area 0 authentication message-digest

network 10.0.0.4 0.0.0.3 area 0

network 10.0.0.8 0.0.0.3 area 0

default-information originate

interface Serial0/0/1

ip ospf message-digest-key 1 md5 cisco123

ip ospf authentication message-digest

```
interface Serial0/1/0
```

```
ip ospf message-digest-key 1 md5 cisco123
```

```
ip ospf authentication message-digest
```

```
! ===== RUTA POR DEFECTO =====
```

```
ip route 0.0.0.0 0.0.0.0 210.10.10.1
```

```
! ===== NAT =====
```

```
ip access-list standard NAT_POOL
```

```
permit 192.168.0.0 0.0.255.255
```

```
permit 172.16.99.0 0.0.0.31
```

```
ip nat inside source list NAT_POOL interface Serial0/0/0 overload
```

```
interface Serial0/0/0
```

```
ip nat outside
```

```
interface Serial0/1/0
```

```
ip nat inside
```

```
interface Serial0/0/1
```

```
ip nat inside
```

```
line vty 0 4
```

```
transport input ssh
```

```
login local
```

```
exec-timeout 5 0
```

```
logging synchronous
```

```
line con 0
```

```
password cisco123
```

```
login
```

logging synchronous

! ===== ACL DE SEGURIDAD =====

ip access-list extended ACL_CORE_IN

remark PERMITIR OSPF (CRUCIAL PARA LA ADYACENCIA)

permit ospf any any

remark BLOQUEAR TELNET (REQUISITO)

deny tcp any any eq 23

remark PERMITIR BACKBONE WAN

permit ip 10.0.0.0 0.255.255.255 any

remark PERMITIR TRÁFICO DE REDES INTERNAS

permit ip 192.168.0.0 0.0.255.255 any

permit ip 172.16.99.0 0.0.0.31 any

permit ip 192.168.200.0 0.0.0.255 any

remark DENEGAR EL RESTO Y LOGUEAR

deny ip any any

interface Serial0/0/1

ip access-group ACL_CORE_IN in

interface Serial0/1/0

ip access-group ACL_CORE_IN in

ip access-list standard ACL_VTY_ACCESS

permit 172.16.99.0 0.0.0.31

deny any

```
line vty 0 4
access-class ACL_VTY_ACCESS in
end
```

Configuración de ISP

```
ISP (Router ISP)
enable
configure terminal
hostname ISP
no ip domain-lookup
banner motd # ADVERTENCIA: Acceso no autorizado esta prohibido. Solo personal autorizado. #

username admin privilege 15 secret cisco123
enable secret cisco123
service password-encryption

ip domain-name empresa.local
crypto key generate rsa general-keys modulus 2048
ip ssh version 2
ip ssh time-out 60
ip ssh authentication-retries 3
login block-for 120 attempts 3 within 60

! ===== INTERFACE WAN =====

interface Serial0/0/0
description ENLACE ISP-R1
ip address 210.10.10.1 255.255.255.252
```

no shutdown

int s0/0/1

description ENLACE ISP-R9

ip address 210.10.11.1 255.255.255.252

no shutdown

ip route 0.0.0.0 0.0.0.0 210.10.10.2

ip route 0.0.0.0 0.0.0.0 210.10.11.2

line vty 0 4

transport input ssh

login local

exec-timeout 5 0

logging synchronous

line con 0

password cisco123

login

logging synchronous

Configuración de la server

R9 (Core - Área 0)

enable

configure terminal

hostname R9

no ip domain-lookup

banner motd # ADVERTENCIA: Acceso restringido. Solo personal autorizado. #

```
username admin privilege 15 secret cisco123
```

```
enable secret cisco123
```

```
service password-encryption
```

```
ip domain-name empresa.local
```

```
crypto key generate rsa general-keys modulus 2048
```

```
ip ssh version 2
```

```
login block-for 120 attempts 3 within 60
```

```
! ===== INTERFACES WAN =====
```

```
interface Serial0/0/0
```

```
description ISP-R9
```

```
ip address 210.10.11.2 255.255.255.252
```

```
no shutdown
```

```
interface g0/1
```

```
description R9-server
```

```
ip address 192.168.200.1 255.255.255.0
```

```
no shutdown
```

```
! ===== RUTA DEFAULT =====
```

```
ip route 0.0.0.0 0.0.0.0 210.10.11.1
```

```
! ===== ACL DE SEGURIDAD =====
```

```
ip access-list extended ACL_SERVER_IN
```

```
remark PERMITIR WEB
```

```
permit tcp any host 192.168.200.2 eq 80
permit tcp any host 192.168.200.2 eq 443
```

```
remark PERMITIR PING
permit icmp any any
```

```
remark BLOQUEAR TELNET
deny tcp any any eq 23
```

```
remark PERMITIR REDES INTERNAS
permit ip 192.168.0.0 0.0.255.255 any
permit ip 172.16.99.0 0.0.0.31 any
permit ip 10.0.0.0 0.255.255.255 any
```

```
remark DENEGAR TODO
deny ip any any
```

```
interface Serial0/0/0
ip access-group ACL_SERVER_IN in
ip access-list standard ACL_VTY_ACCESS
permit 172.16.99.0 0.0.0.31
deny any
```

```
line vty 0 4
access-class ACL_VTY_ACCESS in
transport input ssh
login local
```

exec-timeout 5 0

logging synchronous

line con 0

password cisco123

login

logging synchronous